

Smart India Hackathon 2026 - Presentation Deck Outline (10 Slides)

PS ID: SIH26001 | Ministry of Development of North Eastern Region (MDoNER)

Slide 1: Title Slide

Cloud-Native AI & Physics-Hybrid Landslide Early Warning System. Protecting Critical Himalayan Lifelines across 8 NER States. Live URL: ner-early-warning-system.vercel.app

Slide 2: Regional Context & 4 Failures

Fragile young geology + extreme cloudbursts (>11,000mm/yr). 150+ landslides/yr. Existing systems fail on coarse resolution, storm network blackouts, zero ground truth, and static unranked warnings.

Slide 3: Unique 7-Pillar Solution

1. Physics+GeoAI (FoS + XGBoost/PyTorch) | 2. Citizen Photo & Video Reporting with Auto-GPS | 3. Road-Isolation Graph Intelligence | 4. Offline-First PWA (IndexedDB) | 5. Recharts Weather Nowcast | 6. Continuous Learning Feedback | 7. Digital Twins of NH-10 & SH-5.

Slide 4: End-to-End Enterprise Architecture

Data Layer (Sentinel-1 InSAR, IMD Radar, IoT) -> GeoAI Engine -> PostgreSQL/PostGIS + FastAPI -> Next.js 14 Web PWA + IndexedDB -> 5 Languages + Web Speech Audio Bulletins.

Slide 5: Live Demo Walkthrough Highlights

Tactical GIS with regional camera jumps -> 3.5h advance alerts with 'Listen' voice button -> Video hazard reporting with auto-GPS -> DevTools Offline Mode demonstration proving zero data loss!

Slide 6: Quantitative Impact Benchmarks

3.5 to 6.0 hour lead time vs 0-30 min. 30-meter slope granularity vs 15-25km. Offline local queueing vs system failure. Spoken voice in 5 languages (English, Hindi, Assamese, Bodo, Khasi).

Slide 7: Scalability across 8 NER States

Specific calibrations for Sikkim/Arunachal rockfalls, Meghalaya escarpment saturation, Assam/Tripura railway cuts, and Mizoram urban creep. Zero recurring tile license costs via public Esri Dark Gray canvas.

Slide 8: Tech Stack & Open Standards

Next.js 14, TypeScript, Tailwind CSS, Leaflet, Recharts, FastAPI, PostGIS, XGBoost, PyTorch, OGC GeoJSON, NDMA CAP v1.2, W3C PWA.

Slide 9: Rollout Timeline

Phase 1 (Months 1-2): Pilot staging with SDMA Meghalaya & Sikkim. Phase 2 (Months 3-4): BRO & IMD telemetry hooks. Phase 3 (Months 5-6): National deployment on NIC MeghRaj Cloud.

Slide 10: Conclusion & Q&A

Transforming early warnings from passive post-disaster notifications into actionable, localized, offline-resilient lifelines for North East India. GitHub & Live Vercel Demo.